TODP V1 Architecture Design Document (ADD)

Sections 6–10

Status: FROZEN

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6. Database Architecture

Purpose

Store:

- Opportunities
- Evidence
- Sources
- Citations
- Recommendations
- Audit Logs
- Authority Structures
- Calendar Records
- Registration Records

while maintaining:

- Traceability
- Auditability
- Provenance
- Reproducibility

Core Tables

users

Stores:

- user_id
- profile_metadata
- consent_flags
- location_context

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authorities

Stores:

- authority_id
- authority_name
- authority_type
- parent_authority_id
- source_id
- citation_id
- version

---

opportunities

Stores:

- opportunity_id
- title
- description
- category
- authority_id
- location
- status

---

evidence

Stores:

- evidence_id
- source_id
- citation_id
- retrieval_timestamp
- evidence_type
- version

---

recommendations

Stores:

- recommendation_id
- opportunity_id
- ranking_score
- money_cost

- time_cost
- distance_cost
- version

---

audit_logs

Stores:

- audit_id
- recommendation_id
- audit_type
- audit_status
- timestamp

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calendar_entries

Stores:

- calendar_id
- user_id
- event_id
- sync_status

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registrations

Stores:

- registration_id
- user_id
- opportunity_id
- status

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7. Evidence Store Architecture

Purpose

Evidence is a first-class system object.

No recommendation exists without evidence.

Evidence Model

Evidence

↓

Source

↓

Citation

↓

Timestamp

↓

Recommendation

Evidence Types

Authority Evidence

Examples:

- Government Websites
- University Websites
- NGO Websites
- Official Organization Websites

Opportunity Evidence

Examples:

- Event Pages
- Scholarship Pages
- Course Pages
- Program Pages

Provider Evidence

Examples:

- Official Documentation
- Official Statistics
- Official Reports

Evidence Versioning

Every evidence record must contain:

- version
- retrieval_timestamp
- source_status

Historical versions must remain available.

Retrieval Intelligence Layer

Purpose

Provide evidence retrieval for:

- Opportunities
- Events
- Scholarships
- Government Programs
- Courses
- Authority Definitions
- Institution Information
- Tool Information
- Program Information

Core Components

1. BM25 Retrieval

Purpose:

- Exact Match Search
- Keyword Search

• Authority Search

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2. Vector Retrieval

Purpose:

- Semantic Search
- Context Search
- Similarity Search

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3. Metadata Filtering

Purpose:

Apply retrieval constraints before retrieval.

Discovery Hierarchy:

User Context

↓

Institution

↓

Local

↓

District

↓

State

↓

National

↓

International

Examples:

- state = Jharkhand
- category = scholarship
- type = government_program

---

4. Hybrid Retrieval

Hybrid Score =

BM25 Score

+ Vector Similarity Score

Purpose:

Combine keyword retrieval and semantic retrieval.

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5. Reranking Layer

Purpose:

Improve retrieval quality before recommendation generation.

Inputs:

- BM25 Results
- Vector Results
- Metadata Filtered Results

Outputs:

- Ranked Retrieval Results

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6. Citation Extraction

Purpose:

Extract:

- Sources
- Citations
- Timestamps

for recommendation generation.

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7. Evidence Extraction

Purpose:

Extract evidence objects for:

- Ranking
- Audit
- Provenance
- Recommendation Generation

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8. Audit Architecture

Purpose

Continuously verify:

- Recommendations
- Rankings
- Citations
- Evidence Chains
- Sources

Audit Layers

Source Audit

Verify:

- Source Exists
- Source Reachable
- Source Valid

Citation Audit

Verify:

- Citation Exists

- Citation Valid
- Citation Traceable

Recommendation Audit

Verify:

- Recommendation Traceable
- Recommendation Reproducible

Ranking Audit

Verify:

- Money Cost Correct
- Time Cost Correct
- Distance Cost Correct
- Ranking Score Correct

Retrieval Audit

Verify:

- Retrieval Success
- Retrieval Completeness
- Citation Coverage
- Evidence Coverage

Hallucination Audit

Verify:

- Unsupported Claims
- Missing Sources
- Missing Citations
- Missing Evidence

Core Rule

No Evidence Retrieved

=

No Strong Recommendation

Audit Output

Every audit produces:

- audit_id
- audit_type
- audit_status
- timestamp
- version

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9. API Architecture

Purpose

Provide a consistent interface between:

- Frontend
- Agents
- Database
- External Systems

Core APIs

Discovery API

Functions:

- Search Opportunities
- Search Events
- Search Scholarships
- Search Courses
- Search Programs

Recommendation API

Functions:

- Generate Recommendations
- Retrieve Recommendation Details
- Retrieve Evidence

Authority API

Functions:

- Retrieve Authority Hierarchy
- Retrieve Jurisdiction Hierarchy

Audit API

Functions:

- Retrieve Audit Results
- Retrieve Audit History

Registration API

Functions:

- Autofill Forms
- Retrieve Registration Status

Calendar API

Functions:

- Sync Calendar
- Retrieve Availability
- Create Calendar Entry

API Principles

Every API response must contain:

- source_id
- citation_id
- retrieval_timestamp
- version

when applicable.

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10. Observability Architecture

Purpose

Observe system behavior continuously.

Monitoring Categories

Discovery Monitoring

Track:

- Searches
- Discoveries
- Retrieval Failures

Recommendation Monitoring

Track:

- Recommendations Generated
- Recommendation Acceptance
- Recommendation Failures

Audit Monitoring

Track:

- Audit Passes

- Audit Failures
- Hallucination Events

Source Monitoring

Track:

- Source Availability
- Source Changes
- Source Failures

Agent Monitoring

Track:

- Agent Invocations
- Agent Failures
- Agent Latency

RAG Evaluation Architecture

Purpose

Continuously evaluate retrieval quality.

Evaluation Metrics

- Faithfulness
- Answer Relevance
- Context Precision
- Context Recall
- Citation Accuracy
- Source Coverage
- Recommendation Traceability
- Ranking Reproducibility
- Provenance Completeness

RAG Monitoring

Track:

- Retrieval Success Rate
- Citation Coverage
- Source Coverage
- Missing Citation Rate
- Missing Evidence Rate
- Hallucination Rate
- Ranking Reproducibility Rate
- Recommendation Traceability Rate

Core Questions

Can every recommendation be traced?

Can every ranking be reproduced?

Can every source be audited?

Can every citation be verified?

Can every hallucination be detected?

Can every retrieval be reproduced?

Monitoring Principle

Trace Everything.

Audit Continuously.

Trust Evidence.

Detect Hallucinations.

Preserve Provenance.

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Architecture Classification

TODP V1

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Enterprise RAG

+

Authority Discovery Engine

+

Recommendation Engine

+

Audit Engine

+

Provenance Engine

+

Observability Engine

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END OF FROZEN ADD V1 SECTIONS 6–10